

Software Program Plan - Revision 2

Date: 2026-03-13

Project: Agent Follow-Up

1. Purpose

Deliver a provable follow-up system that ensures background/delegated work triggers proactive operator updates without relying on agent memory or conversational intention.

2. Implementation Gate

Revision 1 was not implementation-ready. Revision 2 closes the pre-implementation blockers by defining: registration interface, observation strategy, task schema, concurrency model, and explicit risk handling.

3. Phase 1 Strategy

- Use a deterministic watchdog.
- Use a SQLite-backed registry.
- Use a scheduler (cron or systemd timer) independent of chat context.
- Use templated notifications, not LLM-backed reasoning, for routine updates.

4. Scope

In Scope

- OpenClaw-agent-initiated background task registration
- Scheduled task observation and reminder logic
- Completion/failure/overdue notifications
- Durable local evidence and recovery after restarts

Out of Scope (Phase 1)

- Arbitrary third-party process introspection without identifiers
- Cross-machine orchestration
- LLM-driven progress summarization as a required control path

5. Risk Register

ID	Risk	Impact	Mitigation
R-1	OpenClaw session/subagent observability is insufficient	High	Phase 1 prioritizes observable process/session-backed tasks only; unsupported types are explicitly labeled.
R-2	Registry corruption or race conditions	High	Use SQLite instead of shared JSON writes.
R-3	Notification spam or duplicates	Medium	Track last notification status and timestamp, plus deterministic suppression logic.
R-4	Scheduler drift or missed runs	Medium	Use cron/systemd logs and late-run detection.
R-5	Overhead from routine evaluation	Medium	Keep Phase 1 non-LLM and rule-based.
R-	Messaging dependency	Medium	Use one concrete OpenClaw-native emitter path and retain

6. Milestones

1. Approve Revision 2 design package.
2. Build registration and storage layer.
3. Build watchdog and scheduler integration.
4. Build notification emission path.
5. Execute V&V plan and produce proof package.

7. Success Criteria

- No manual status request required for tracked tasks.
- At least one completion, one overdue reminder, and one failure path are proven.
- System survives agent/session restart with tracked task state intact.